

Direct Generation of a Somos–4 Sequence from an Algebraic Generating Function

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Abstract

We record a five-parameter quadratic algebraic generating function whose coefficient sequence has prescribed initial Hankel determinants (λ, m, r, η) and whose Hankel transform lies in the Somos–4 family $A(1, \tau)$. We begin by introducing the Hankel data and the corresponding parameters. We then describe the underlying Stieltjes continued-fraction construction, its normalization, and the coefficient recurrence, before writing down the explicit algebraic equation. After that we discuss the two square-root branches, their coalescence at the origin, a desingularized coefficient recurrence, the induced shift in the Hankel parameters, and the nonuniqueness of the underlying coefficient-sequence representation. The essential point is that the construction removes this ambiguity by prescribing, in addition to the ordinary Hankel transform, a companion Hankel sequence obtained from the same Somos–4 orbit by a two-step shift.

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1 Context and relation to earlier work

The relation between Hankel determinants, continued fractions, moment sequences, and nonlinear recurrences has a long history. Classical continued-fraction treatments include Wall [19] and Jones and Thron [12]; the combinatorial theory of continued fractions was developed systematically by Flajolet [5]. Determinantal evaluations and transformations relevant to Hankel matrices are discussed, for example, by Krattenthaler [13], Layman [14], and French [7]. The quadratic transformation of Sulanke and Xin [17] and related continued-fraction methods of Gessel and Xin [9] have become particularly effective tools for proving structured Hankel-determinant identities.

Somos recurrences came from the work of Somos and were made widely known by Gale [8]. Their Laurent phenomenon and their connections with elliptic divisibility sequences and elliptic curves have been discussed by many authors; see, for example, Fomin and Zelevinsky [6], van der Poorten and Swart [18], Hone [10], and Hone and Swart [11]. Xin proved the original Somos–4 Hankel determinant conjecture by a continued-fraction recursion system [21].

A substantial body of work by Barry studies generating functions whose Hankel transforms satisfy Somos-type recurrences. This includes generalized Catalan recurrences [1], generalized Narayana triangles [2], Riordan–Bernstein constructions [3], and Riordan pseudo-involutions, continued fractions, and elliptic curves [4]. Barry’s examples and conjectures form an important part of the immediate background of the present construction.

Wang and Zhang subsequently gave a sufficient condition for an algebraic generating function to have an (α, β) Somos–4 Hankel transform, using the Sulanke–Xin machinery [20]. Liu, Wang, and Zhang then extended this condition over a rational-function field and used it to settle the

outstanding Barry conjectures treated there; they also studied odd and even Somos subsequences, the Hurwitz transform, orthogonal-polynomial Hankel formulas, periodic Hankel determinants, and elliptic parameter families [15].

The point of view here is different. Instead of beginning with a given generating function and deriving the parameters of its Hankel recurrence, we start with prescribed initial Hankel data (λ, m, r, η) and the Somos coefficient τ , impose a companion Hankel orbit, and construct an explicit quadratic algebraic generating function realizing those data. The two-step companion prescription is what selects one coefficient sequence from among the generally nonunique sequences having the same ordinary Hankel transform.

2 Hankel determinants and Somos parameters

Let

$$G(z) = \sum_{n \geq 0} g_n z^n$$

be a formal power series, and for $n \geq 1$ define its ordinary Hankel determinants by

$$H_n = \det(g_{i+j})_{0 \leq i, j \leq n-1}, \quad H_0 = 1. \quad (1)$$

The five parameters $\lambda, m, r, \eta, \tau$ have the following roles:

$$H_1 = \lambda, \quad H_2 = m, \quad H_3 = r, \quad H_4 = \eta, \quad (2)$$

and the Hankel sequence satisfies the Somos-4 recurrence

$$H_{n+4}H_n = H_{n+3}H_{n+1} + \tau H_{n+2}^2. \quad (3)$$

Thus τ is the second coefficient of the Somos family $A(1, \tau)$, whereas λ, m, r , and η select the initial Hankel determinants. The algebraic generating function producing these determinants is given in the next section. The construction makes one additional choice that is easy to overlook: together with $(H_n)_{n \geq 0}$, it fixes the companion Hankel sequence

$$\hat{H}_n = \det(g_{i+j+1})_{0 \leq i, j \leq n-1} = H_{n+2} \quad (n \geq 1). \quad (4)$$

Thus the companion is obtained by advancing the Somos-4 orbit by two indices without applying any additional normalization factor. Its first term is $\hat{H}_1 = g_1 = H_3 = r$, not $H_2 = m$. This second prescription is part of the definition of the selected coefficient sequence; it is not a consequence of the ordinary Hankel determinants alone. Section 8 explains why prescribing both sequences removes the degree of freedom discussed there.

3 Behind the construction: the Stieltjes fraction

The quadratic equation below may be obtained by first constructing a Stieltjes continued fraction whose Hankel determinants follow the prescribed Somos-4 orbit. Write

$$G(x) = \frac{c_0}{1 - \frac{c_1 x}{1 - \frac{c_2 x}{1 - \frac{c_3 x}{\ddots}}}}. \quad (5)$$

For this normalization, the ordinary Hankel determinants are [19, 12, 5]

$$H_n = c_0^n \prod_{j=1}^{n-1} (c_{2j-1} c_{2j})^{n-j}. \quad (6)$$

Thus

$$H_1 = c_0, \quad H_2 = c_0^2 c_1 c_2, \quad H_3 = c_0^3 c_1^2 c_2^2 c_3 c_4,$$

and similarly at higher orders.

It is useful first to remove the exponential scaling caused by H_1 . Set

$$a_1 = 1, \quad a_2 = \frac{m}{\lambda^2}, \quad a_3 = \frac{r}{\lambda^3}, \quad a_4 = \frac{\eta}{\lambda^4}. \quad (7)$$

Then $H_n = \lambda^n a_n$ for $1 \leq n \leq 4$, and the normalized sequence (a_n) obeys the same Somos-4 recurrence,

$$a_{n+4} a_n = a_{n+3} a_{n+1} + \tau a_{n+2}^2. \quad (8)$$

The powers of λ in (7) are essential: for nonunit $H_1 = \lambda$, the correct normalization is $a_j = H_j / \lambda^j$, not H_j / λ^{j-1} .

A convenient initial segment of the Stieltjes coefficients is

$$c_0 = \lambda, \quad c_1 = \lambda^2 a_3, \quad c_2 = \frac{a_2}{\lambda^2 a_3}, \quad (9)$$

$$c_3 = \frac{a_4}{a_2 a_3}, \quad c_4 = \frac{a_3^2}{a_2 a_4}, \quad c_5 = \tau \frac{a_2 a_3}{a_4} + \frac{a_2^2}{a_3}. \quad (10)$$

Equivalently, in the parameters used throughout this paper,

$$\begin{aligned} c_0 &= \lambda, & c_1 &= \frac{r}{\lambda}, & c_2 &= \frac{m}{\lambda r}, \\ c_3 &= \frac{\lambda \eta}{mr}, & c_4 &= \frac{r^2}{m\eta}, & c_5 &= \frac{\tau mr}{\lambda \eta} + \frac{m^2}{\lambda r}. \end{aligned} \quad (11)$$

The remaining coefficients are generated alternately by

$$c_{2k} = \frac{1}{c_{2k-2} c_{2k-1}}, \quad c_{2k+1} = c_{2k-2} + \frac{\tau}{c_{2k-1}} \quad (k \geq 3). \quad (12)$$

In particular,

$$c_6 = \frac{1}{c_4 c_5}, \quad c_7 = c_4 + \frac{\tau}{c_5}, \quad c_8 = \frac{1}{c_6 c_7}, \quad c_9 = c_6 + \frac{\tau}{c_7}.$$

Substitution in (6) gives $H_j = \lambda^j a_j$ for $1 \leq j \leq 4$. At the next order it gives

$$\frac{H_5}{\lambda^5} = \frac{a_3 a_4 c_5}{a_2} = a_2 a_4 + \tau a_3^2,$$

which is exactly the first Somos-4 step. The alternating recurrence (12) continues this mechanism.

The initial coefficients in (10) do more than reproduce the ordinary Hankel determinants. The standard shifted-Hankel product formula for the Stieltjes fraction gives

$$\det(g_{i+j+1})_{0 \leq i, j \leq n-1} = H_{n+2} \quad (n \geq 1). \quad (13)$$

Indeed, the first case is

$$g_1 = c_0 c_1 = \lambda \frac{r}{\lambda} = r.$$

Hence the second product sequence attached to the Stieltjes fraction is the two-step companion in (4). The ordinary Hankel transform determines only the products $c_{2j-1}c_{2j}$ in (6); the companion sequence fixes the corresponding splitting into the odd and even Stieltjes coefficients. Consequently, once both product sequences and the normalization $c_0 = \lambda$ are prescribed, the Stieltjes fraction, and hence the selected coefficient sequence, is fully determined on the nondegenerate locus. This is the particular representative used throughout the paper. The next step is to eliminate the recursively defined continued-fraction tail. The following isolated calculation explains why this elimination closes quadratically and hence provides the conceptual bridge to the explicit equation in Section 4.

3.1 Quadratic closure of the recurrent tail

For a sequence $(d_n)_{n \geq 1}$, write

$$T_n(x) = \frac{1}{1 - \frac{d_n x}{1 - \frac{d_{n+1} x}{1 - \ddots}}}, \quad T_n = \frac{1}{1 - d_n x T_{n+1}}. \quad (14)$$

The recurrence in (12) becomes transparent after the tail is started at c_4 . Put

$$d_n = c_{n+3}, \quad b = d_1 = c_4, \quad d = d_2 = c_5, \quad t = \tau. \quad (15)$$

Then $d_1 = b$, $d_2 = d$, $d_3 = 1/(bd)$, and thereafter

$$d_n = \frac{1}{d_{n-1}d_{n-2}} \quad (n \geq 3 \text{ odd}), \quad d_n = \frac{t}{d_{n-2}} + d_{n-3} \quad (n \geq 4 \text{ even}). \quad (16)$$

Thus the alternating recurrence can be studied independently of the first three levels of the original Stieltjes fraction.

Proposition 1 (Quadratic equation for the recurrent tail). *Let $T = T_1$ be the formal continued fraction defined by (14)–(16). Then $T(0) = 1$ and*

$$\mathcal{A}(x)T^2 + \mathcal{B}(x)T + \mathcal{C}(x) = 0, \quad (17)$$

where

$$\mathcal{A}(x) = (bt + 1)x - b((b + d)t + 1)x^2, \quad (18)$$

$$\mathcal{B}(x) = bdt x^2 + (-bt + b^2d + d^2b - 1)x - bd, \quad (19)$$

$$\mathcal{C}(x) = bd - bd^2x. \quad (20)$$

Proof. Removing one level of a Stieltjes fraction is a fractional-linear transformation,

$$T_{n+1} = \frac{T_n - 1}{d_n x T_n}. \quad (21)$$

Consequently, quadratic relations are preserved under tail extraction. More explicitly, if $Q_n(Y)$ is quadratic and $Q_n(T_n) = 0$, then a polynomial for the next tail is

$$Q_{n+1}(Y) = (1 - d_n x Y)^2 Q_n\left(\frac{1}{1 - d_n x Y}\right), \quad (22)$$

after removal of any nonzero common factor. Substitution of (16) into successive applications of (22) uses only

$$d_n d_{n-1} d_{n-2} = 1 \quad (n \text{ odd}), \quad d_n d_{n-2} = t + d_{n-3} d_{n-2} \quad (n \text{ even}). \quad (23)$$

On collecting the coefficients of Y^2 , Y , and 1 , these identities cancel the newly introduced tail coefficient and return the same quadratic family. Initializing with $d_1 = b$, $d_2 = d$, and $d_3 = 1/(bd)$ gives (18)–(20). At $x = 0$, (17) reduces to $bd(1 - T) = 0$, so the continued-fraction branch is the branch with $T(0) = 1$. $\square$

This proposition gives a short derivation of the algebraic nature of the full generating function. Let $T = T_4$ denote the tail of (5) beginning with c_4 . The first three levels are recovered by

$$T_3 = \frac{1}{1 - c_3 x T}, \quad T_2 = \frac{1}{1 - c_2 x T_3}, \quad T_1 = \frac{1}{1 - c_1 x T_2}, \quad G = c_0 T_1. \quad (24)$$

Each step is again a Möbius transformation. Substituting (24) into (17), followed by $c_0 = \lambda$ and the values in (11), therefore produces a quadratic polynomial in G . Clearing denominators and collecting powers of z yields exactly the polynomials in (28)–(36). This explains structurally why the recursively defined infinite fraction gives a quadratic algebraic generating function: the recurrence is the compatibility condition that keeps the continued-fraction tails inside a quadratic family, while the omitted initial levels only conjugate that family by fractional-linear maps.

4 The algebraic generating function

Define the polynomials $A(z)$, $B(z)$, $C(z)$ by

$$A(z) = \lambda \eta m r^2 (\tau - \eta) z^3 \quad (25)$$

$$+ r \left(-(\lambda^2 \eta m + m^2(m + r^2))\tau + (2m - 1)\lambda^2 \eta^2 + \eta m^2(m + r^2) - \lambda r(m + r^2) \right) z^2 \quad (26)$$

$$+ \lambda \left(m^2 r^2 \tau - (m - 1)\lambda^2 \eta^2 - m(m^2 + 2mr^2 - m - r^2)\eta + \lambda r^3 \right) z \quad (27)$$

$$+ \lambda^2 \eta m(m - 1)r, \quad (28)$$

$$B(z) = -\lambda^2 \eta m r^2 \tau z^3 \quad (29)$$

$$+ r \left((2\lambda^3 \eta m + \lambda m^2(2m + r^2))\tau - (2m - 1)\lambda^3 \eta^2 - \lambda \eta m^3 + \lambda^2 r(2m + r^2) \right) z^2 \quad (30)$$

$$+ \lambda^2 \left(-2m^2 r^2 \tau + 2(m - 1)\lambda^2 \eta^2 + m(2m^2 + 2mr^2 - 2m - r^2)\eta - 2\lambda r^3 \right) z \quad (31)$$

$$- 2\lambda^3 \eta m(m - 1)r, \quad (32)$$

$$C(z) = -\lambda^2 m r \left(\tau(\lambda^2 \eta + m^2) + \lambda r \right) z^2 \quad (33)$$

$$+ \left(\lambda^3 m^2 r^2 \tau + \lambda^4 r^3 + \eta(m - 1)(\eta(-15\lambda^4 + 85\lambda^3 - 225\lambda^2 + 274\lambda - 120) \right. \quad (34)$$

$$\left. - \lambda^3 m^2 - 3(\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4)(\lambda - 5) \right) z \quad (35)$$

$$+ \lambda^4 \eta m(m - 1)r. \quad (36)$$

The generating function is the formal-power-series branch of

$$A(z)G(z)^2 + B(z)G(z) + C(z) = 0 \quad (37)$$

selected by the initial conditions implicit in (2), beginning with $g_0 = \lambda$ and $g_1 = r$.

Remark 1 (The apparently mysterious integer coefficients). The polynomial occurring in (36) is simply

$$-15\lambda^4 + 85\lambda^3 - 225\lambda^2 + 274\lambda - 120 = \frac{6!}{\lambda} \binom{\lambda}{6} - \lambda^5 \quad (38)$$

$$= (\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4)(\lambda - 5) - \lambda^5. \quad (39)$$

Thus the coefficients $-15, 85, -225, 274, -120$ arise by removing the leading term λ^5 from the falling factorial $(\lambda - 1) \cdots (\lambda - 5)$.

5 The two algebraic branches

With

$$D(z) = B(z)^2 - 4A(z)C(z),$$

the two algebraic branches may be represented by

$$G(z) = \frac{-B(z) \pm \sqrt{D(z)}}{2A(z)}. \quad (40)$$

When the constant term of A vanishes, the rationalized expressions

$$\frac{-2C(z)}{B(z) \mp \sqrt{D(z)}}$$

can be used instead. The sign in (40) cannot be fixed uniformly over the full parameter space.

Write

$$P(z, X) = A(z)X^2 + B(z)X + C(z).$$

The constant coefficients of (28)–(36) are

$$A_0 = \lambda^2 \eta m(m-1)r, \quad B_0 = -2\lambda^3 \eta m(m-1)r, \quad C_0 = \lambda^4 \eta m(m-1)r.$$

Consequently,

$$P(0, X) = A_0(X - \lambda)^2. \quad (41)$$

Both algebraic branches therefore coalesce at $X = \lambda$ when $z = 0$, and

$$\partial_X P(0, \lambda) = 2A_0\lambda + B_0 = 0. \quad (42)$$

The simple-root form of the formal implicit-function theorem is not applicable at the origin. The desired branch is instead selected by expanding the square root and requiring the prescribed coefficients, beginning with $g_0 = \lambda$ and $g_1 = r$.

Set

$$G(z) = \lambda + zF(z), \quad F(z) = \sum_{q \geq 0} f_q z^q, \quad f_q = g_{q+1}. \quad (43)$$

For

$$D_j = 2\lambda A_j + B_j, \quad E_j = \lambda^2 A_j + \lambda B_j + C_j,$$

where $C_3 = 0$, substitution into $P(z, G(z)) = 0$ gives

$$P(z, \lambda + zF) = \sum_{j=0}^3 E_j z^j + z \left(\sum_{j=0}^3 D_j z^j \right) F + z^2 \left(\sum_{j=0}^3 A_j z^j \right) F^2.$$

Equation (41) gives $E_0 = D_0 = 0$. An ordinary power-series branch additionally requires

$$E_1 = 0. \quad (44)$$

The next coefficient equation is

$$A_0 f_0^2 + D_1 f_0 + E_2 = 0. \quad (45)$$

Since $f_0 = g_1 = r$, the prescribed value of r selects one of the two branches of the desingularized quadratic.

6 The two-step companion

Let $G_{0;\ell,m,r,n,t}(x) = \sum_{q \geq 0} g_q x^q$ denote the selected solution. Its Hankel determinants satisfy

$$(H_1, H_2, H_3, H_4) = (\ell, m, r, n), \quad H_{q+4}H_q = H_{q+3}H_{q+1} + tH_{q+2}^2. \quad (46)$$

The next two values are

$$H_5 = \frac{mn + tr^2}{\ell}, \quad (47)$$

$$H_6 = \frac{rH_5 + tn^2}{m} = \frac{r(mn + tr^2) + t\ell n^2}{\ell m}. \quad (48)$$

Remove the first coefficient and shift the remaining series to the origin:

$$\hat{G}(x) = \frac{G_{0;\ell,m,r,n,t}(x) - g_0}{x} = \sum_{q \geq 0} g_{q+1} x^q. \quad (49)$$

Its Hankel transform is

$$\det(g_{i+j+1})_{0 \leq i,j \leq q-1} = H_{q+2} \quad (q \geq 1). \quad (50)$$

It therefore begins with

$$r, \quad n, \quad \frac{mn + tr^2}{\ell}, \quad \frac{r(mn + tr^2) + t\ell n^2}{\ell m}, \dots$$

This is exactly the two-step advanced orbit $(H_{q+2})_{q \geq 1}$. It therefore satisfies the same recurrence with parameter t . Equivalently, it has the initial parameter tuple

$$\left(r, n, \frac{mn + tr^2}{\ell}, \frac{r(mn + tr^2) + t\ell n^2}{\ell m}, t \right). \quad (51)$$

The assertion is an identity of Hankel transforms, not necessarily of the underlying coefficient sequences.

7 A recurrence for the coefficients

Using the notation of (43), division of the expanded quadratic equation by z^2 yields

$$A(z)F(z)^2 + (D_1 + D_2z + D_3z^2)F(z) + E_2 + E_3z = 0. \quad (52)$$

The branch condition is

$$A_0r^2 + D_1r + E_2 = 0, \quad (53)$$

and the branch is simple precisely when

$$2A_0r + D_1 \neq 0. \quad (54)$$

Equivalently,

$$(2A_0r + D_1)^2 = D_1^2 - 4A_0E_2. \quad (55)$$

Define

$$T_q = \sum_{k=0}^q f_k f_{q-k}, \quad T_q = 0 \quad (q < 0). \quad (56)$$

Coefficient extraction from (52) gives

$$\sum_{j=0}^3 A_j T_{q-j} + D_1 f_q + D_2 f_{q-1} + D_3 f_{q-2} + E_{q+2} = 0, \quad (57)$$

where $f_q = 0$ for $q < 0$ and $E_j = 0$ for $j > 3$. For $q \geq 1$,

$$T_q = 2r f_q + \sum_{k=1}^{q-1} f_k f_{q-k}. \quad (58)$$

Under condition (54), the coefficients are therefore generated recursively by

$$f_q = - \frac{A_0 \sum_{k=1}^{q-1} f_k f_{q-k} + \sum_{j=1}^3 A_j T_{q-j} + D_2 f_{q-1} + D_3 f_{q-2} + E_{q+2}}{2A_0 r + D_1}. \quad (59)$$

This quadratic convolution recurrence follows directly from the algebraic generating-function equation. On the exceptional locus $2A_0 r + D_1 = 0$, the two desingularized branches coalesce at the origin and the next coefficient is not determined by the first-order coefficient equation; higher-order compatibility or a Puiseux expansion may then be required.

Because F is algebraic of degree at most two, it is D-finite. Consequently, the same coefficient sequence also satisfies a fixed-width linear recurrence with polynomial coefficients in the index.

8 Nonuniqueness of the coefficient sequence

The ordinary Hankel transform does not, in general, determine the coefficient sequence. If only (H_n) is prescribed, the factorization $c_{2j-1}c_{2j}$ visible in (6) leaves room for variation in the individual Stieltjes coefficients. For this reason the five-parameter algebraic generating function should not be read as the unique moment representation attached to an ordinary Somos-4 Hankel orbit.

The construction in this paper removes this freedom by imposing the additional companion condition (4). In other words, we prescribe two product sequences: the first is the Somos-4 sequence (H_n) , and the second is the same sequence advanced by two determinant indices. Together with $c_0 = \lambda$, these two prescriptions determine the odd and even Stieltjes coefficients separately through (10) and (12). Hence the algebraic generating function in Section 4 is the fully specified representative selected by those two Hankel-level requirements. This choice is already built into the initial Stieltjes coefficients above; making it explicit here clarifies how the apparent degree of freedom is eliminated.

Other moment sequences can still share the first, ordinary Hankel transform when the companion condition is not imposed. For example, the binomial transform, a standard Hankel-transform-preserving operation [14, 7],

$$g'_q = \sum_{k=0}^q \binom{q}{k} g_k \quad (60)$$

has generating function

$$G'(z) = \frac{1}{1-z} G\left(\frac{z}{1-z}\right)$$

and preserves the Hankel determinants. Scalar and exponential rescalings of coefficient sequences produce explicit normalization factors in the determinants, while affine changes of the underlying variable give further related moment functionals. Equality of Hankel transforms must therefore not be replaced by equality of generating functions without an independent argument.

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